

AtLease: Decentralized Warehouse Leasing Certificates Using Blockchain

Mrs. Abhilasha Kore, Abhishek Biradar, Purvesh Jambhulkar, Viraj Gudhetwar, Piyush Mogare
Department of Information Technology
D. Y. Patil College of Engineering
India

Abstract—Traditional warehouse leasing systems rely on centralized databases or paper-based certificates, which are vulnerable to tampering, loss, and ownership disputes. These limitations result in reduced transparency and increased dependence on intermediaries for verification. To address these challenges, this paper proposes *AtLease*, a decentralized warehouse leasing certificate system built on blockchain technology. The system represents each lease agreement as a non-transferable NFT, implemented using a soulbound token mechanism to ensure that certificates remain permanently bound to the tenant's wallet. Smart contracts written in Solidity are used to mint and manage lease certificates, while decentralized storage via IPFS is employed to store lease metadata securely and cost-effectively. The proposed architecture integrates a React-based frontend, a Node.js blockchain service layer, IPFS for metadata storage, and the Polygon blockchain for deployment. Transfer functionality is disabled at the smart contract level to prevent misuse or resale of lease certificates. Experimental results demonstrate successful certificate issuance, public verifiability through blockchain explorers, and immutable metadata storage. *AtLease* provides a tamper-proof, transparent, and trust-minimized approach to warehouse leasing, illustrating how blockchain can be applied to real-world certification use cases beyond financial applications.

Index Terms—Blockchain, NFT, Soulbound Tokens, Smart Contracts, IPFS, Warehouse Leasing

I. INTRODUCTION

Traditional warehouse leasing systems typically rely on centralized digital platforms or paper-based documentation to manage lease agreements and ownership records. Such approaches suffer from several limitations, including vulnerability to data tampering, document loss, and lack of transparent verification mechanisms. In many cases, tenants and property owners must depend on intermediaries to validate lease certificates, which increases operational overhead and introduces trust-related challenges. These issues become more significant in large-scale logistics environments where multiple stakeholders interact across organizational boundaries.

Blockchain technology offers a decentralized and immutable data layer that can address these limitations by providing a single, verifiable source of truth [1], [2]. By recording transactions on a distributed ledger, blockchain eliminates reliance on centralized authorities while ensuring transparency and tamper resistance. However, most existing blockchain-based solutions primarily focus on financial applications or transferable digital assets [3]. In real-world leasing scenarios, certificates should not be tradable, as ownership transfer can lead to misuse or fraudulent representation.

To address this gap, this paper presents *AtLease*, a decentralized warehouse leasing certificate system that utilizes non-transferable NFTs implemented through a soulbound token mechanism. Each lease agreement is represented as a unique digital certificate permanently bound to the tenant's wallet, preventing resale or unauthorized transfer. The proposed system combines smart contracts written in Solidity, decentralized storage via IPFS, and deployment on the Polygon blockchain. A React-based frontend and Node.js service layer enable user interaction and blockchain communication. The primary contribution of this work is the design and implementation of a blockchain-based certification framework that emphasizes immutability, ownership authenticity, and practical deployment for warehouse leasing use cases.

II. RELATED WORK

Several studies have explored the application of blockchain technology in property management and digital certification systems [4], [7]. Existing research demonstrates the use of distributed ledgers to improve transparency and reduce fraud in land registry and real estate ownership records. These systems leverage blockchain immutability to provide verifiable proof of ownership while minimizing reliance on centralized authorities.

Recent advancements have also introduced non-fungible tokens (NFTs) as digital representations of unique assets [6]. While NFTs offer uniqueness and traceability, most implementations allow unrestricted transferability, making them unsuitable for certification scenarios where ownership must remain fixed. Transferable NFTs can lead to misuse, unauthorized resale, and ambiguity in legal contexts.

Decentralized storage solutions such as IPFS have been widely adopted to overcome the cost limitations of on-chain data storage [5]. Several blockchain-based platforms combine smart contracts with IPFS to manage digital documents efficiently. However, many existing systems focus primarily on financial tokenization rather than practical certification workflows.

Although prior work highlights the potential of blockchain for secure record management, limited attention has been given to enforcing non-transferable ownership at the protocol level for leasing applications. The proposed *AtLease* system addresses this gap by integrating soulbound token behavior directly into the smart contract, ensuring that lease certificates

remain permanently bound to tenant wallets. This approach enables tamper-proof certification while preserving decentralization and practical deployability for warehouse leasing scenarios.

III. PROBLEM STATEMENT

Current warehouse leasing processes predominantly rely on centralized databases or physical documentation to manage lease agreements and ownership verification. Such systems are vulnerable to unauthorized modification, document loss, and operational inconsistencies caused by reliance on trusted intermediaries. In dispute scenarios, tenants and property owners often lack access to a single tamper-proof source of truth, leading to prolonged verification procedures and reduced transparency.

While blockchain technology provides immutability and decentralized verification, existing implementations largely utilize transferable digital assets, which are unsuitable for lease certification. Standard NFT-based systems permit ownership transfer, enabling potential misuse, resale, or fraudulent re-assignment of lease certificates. Furthermore, storing complete lease data directly on-chain is economically impractical due to transaction costs and scalability constraints.

Therefore, a system is required that provides immutable lease certification while preventing unauthorized ownership transfer and minimizing on-chain storage overhead. Such a solution must ensure permanent binding of certificates to tenant identities, support decentralized verification, and maintain practical deployability. Addressing these requirements forms the basis of the proposed AtLease framework.

IV. PROPOSED SYSTEM ARCHITECTURE

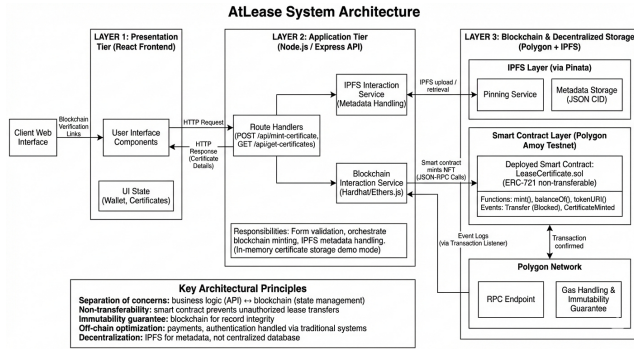


Fig. 1. AtLease System Architecture

The overall architecture of the AtLease system is illustrated in Fig. 1. The proposed framework follows a three-layered design consisting of a presentation tier, an application tier, and a blockchain and decentralized storage tier. This layered approach ensures separation of concerns, modularity, and scalability while maintaining secure interaction with blockchain components.

The presentation tier is implemented as a React-based web interface that enables users to browse warehouses, initiate lease requests, and view issued certificates. This layer manages

client-side state, including wallet connectivity and certificate visualization, and communicates with the backend through standard HTTP requests.

The application tier is built using a Node.js and Express-based REST API server, which acts as an orchestration layer between the frontend and blockchain infrastructure. It handles form validation, request routing, and coordination of certificate minting operations. Upon receiving lease data, the backend invokes the IPFS interaction service to upload structured lease metadata in JSON format. The resulting content identifier (CID) is returned and subsequently passed to the blockchain interaction service, which utilizes Ethers.js to communicate with the deployed smart contract through JSON-RPC calls.

The blockchain and decentralized storage tier consists of the Polygon network and IPFS, integrated via a pinning service. Lease metadata is stored off-chain on IPFS to reduce on-chain storage costs, while the corresponding CID is recorded on-chain as the tokenURI of the NFT. The smart contract, implemented using the ERC-721 standard, mints a unique certificate for each lease agreement. Transfer functionality is explicitly disabled at the contract level to enforce soulbound behavior, ensuring that certificates remain permanently bound to tenant wallets.

Transaction confirmations and contract events are monitored by the backend through event listeners, enabling real-time updates to the frontend. This design guarantees immutability of lease records, decentralized verification, and prevention of unauthorized ownership transfer. Furthermore, the architecture isolates blockchain-specific logic from application workflows, allowing future extensions such as payment integration and backend verification without modifying the core smart contract layer.

V. METHODOLOGY AND IMPLEMENTATION

The AtLease system is implemented using a modular approach that integrates smart contracts, decentralized storage, and a web-based user interface. The methodology focuses on enforcing non-transferable lease certificates, minimizing on-chain storage costs, and maintaining a clear separation between application logic and blockchain operations.

A. Smart Contract Design

The core of the system is a Solidity-based smart contract named *LeaseCertificate.sol*, developed using the ERC-721 standard. Each lease agreement is represented as a unique NFT minted by the platform-controlled wallet. The contract owner is responsible for issuing certificates, ensuring controlled minting and preventing unauthorized creation of lease records. The tokenURI associated with each NFT stores a reference to off-chain metadata hosted on IPFS, enabling lightweight on-chain storage while preserving data integrity.

The contract exposes essential functions such as `mint()`, `balanceOf()`, and `tokenURI()`, allowing certificate creation and verification. Contract events are emitted upon successful minting, enabling backend services to monitor transaction completion and synchronize application state.

B. Soulbound Token Enforcement

Standard ERC-721 tokens permit ownership transfer, which is unsuitable for lease certification. To address this limitation, AtLease implements soulbound behavior by overriding transfer-related functions within the smart contract. Any attempt to transfer a certificate results in transaction reversion, thereby enforcing non-transferability directly at the protocol level. This ensures that lease certificates remain permanently bound to tenant wallets and cannot be resold or reassigned, even through external blockchain interactions.

C. Metadata Storage Using IPFS

Storing complete lease information directly on-chain is economically inefficient. Therefore, AtLease adopts a hybrid storage model in which structured lease metadata is stored off-chain using IPFS. The metadata includes warehouse identifiers, lease duration, lessee reference, and issuer information, formatted as a JSON document. This file is uploaded through a pinning service, which returns a content identifier (CID). The CID is then recorded on-chain as the tokenURI, enabling decentralized access and tamper detection through cryptographic hashing.

D. Frontend Integration

The frontend is implemented as a React-based single-page application that provides interfaces for warehouse selection, lease initiation, and certificate viewing. User actions trigger REST API calls to the backend, which coordinates IPFS uploads and smart contract interactions. Wallet connectivity enables users to view issued certificates and verify ownership. Transaction confirmations and event logs are processed by the backend and reflected in the user interface, providing real-time feedback during certificate issuance.

VI. SYSTEM WORKFLOW

The operational workflow of the AtLease system follows a sequential process that coordinates frontend interaction, backend orchestration, decentralized storage, and blockchain execution.

Initially, the user selects a warehouse through the web interface and initiates a lease request. The frontend collects the required lease parameters and forwards them to the backend via RESTful API calls. Upon receiving the request, the backend validates the input and constructs structured lease metadata in JSON format.

Next, the metadata is uploaded to IPFS through a pinning service, which returns a unique content identifier (CID). This CID serves as a permanent reference to the lease document and is passed to the blockchain interaction module. The backend then invokes the smart contract's mint() function, supplying the tenant wallet address and the generated tokenURI.

Once the transaction is confirmed on the Polygon blockchain, a non-transferable NFT certificate is issued and bound to the tenant's wallet. Smart contract events are monitored by the backend to detect successful minting. The frontend is subsequently updated to display the issued certificate, including the token identifier and IPFS metadata link.

For demonstration purposes, payment verification is excluded from the current implementation. In a production environment, the same workflow would be triggered after successful payment confirmation. This end-to-end process ensures immutable certificate issuance, decentralized verification, and permanent ownership binding.

VII. SECURITY CONSIDERATIONS

Security is a primary design objective of the AtLease framework. The system ensures that private keys are never exposed to the frontend, and all blockchain transactions are initiated through secure backend services. Certificate minting is restricted to the platform-controlled wallet, preventing unauthorized creation of lease records.

Sensitive personal information is not stored on-chain. Only IPFS content identifiers are recorded in the smart contract, while detailed lease metadata remains off-chain, reducing privacy risks and minimizing blockchain storage costs. Wallet ownership serves as cryptographic proof of certificate authenticity, eliminating the need for centralized verification authorities.

Non-transferability is enforced directly at the smart contract level by overriding ERC-721 transfer functions, ensuring that certificates cannot be reassigned or resold, even through external blockchain tools. This protocol-level restriction prevents misuse and maintains the integrity of lease ownership.

Additionally, blockchain immutability guarantees that once a lease certificate is issued, it cannot be altered or deleted. This design provides tamper resistance, transparent verification, and long-term availability of records while maintaining a privacy-conscious architecture suitable for real-world leasing applications.

VIII. RESULTS AND DISCUSSION

The proposed AtLease system was successfully implemented and evaluated in a controlled development environment. Lease certificates were minted on the Polygon blockchain and issued as non-transferable NFTs bound to tenant wallets. Each certificate contains a unique token identifier and an associated tokenURI referencing metadata stored on IPFS.

Experimental results demonstrate that lease metadata uploaded to IPFS remains immutable, with any modification resulting in a different content identifier, thereby enabling tamper detection. Smart contract deployment and minting operations were verified through blockchain explorers, confirming successful transaction execution and permanent on-chain record creation.

The soulbound enforcement mechanism effectively prevented certificate transfer attempts, validating that ownership reassignment is restricted at the protocol level rather than through application logic. Users were able to view issued certificates directly in their wallets and retrieve metadata via the IPFS link, ensuring transparent and decentralized verification.

From a system performance perspective, off-chain storage significantly reduced blockchain transaction costs while

maintaining data integrity. The modular architecture enabled seamless coordination between frontend interaction, backend orchestration, decentralized storage, and smart contract execution. Although payment integration was excluded from the demonstration, the workflow supports straightforward extension for production deployment.

These results confirm that AtLease provides a practical and secure framework for decentralized lease certification, combining immutability, ownership authenticity, and usability without introducing unnecessary blockchain overhead.

IX. CONCLUSION AND FUTURE SCOPE

This paper presented AtLease, a decentralized warehouse leasing certificate system that utilizes blockchain technology to provide immutable and verifiable lease records. By representing lease agreements as non-transferable NFTs through a soulbound token mechanism, the proposed framework ensures permanent ownership binding while preventing unauthorized transfer or resale. The integration of smart contracts, IPFS-based metadata storage, and a modular application architecture demonstrates a practical approach to applying blockchain for real-world certification rather than financial asset trading.

The implemented system successfully achieved decentralized certificate issuance, transparent verification, and protocol-level enforcement of non-transferability. Off-chain metadata storage minimized blockchain overhead while preserving data integrity, and wallet-based ownership provided cryptographic authenticity without reliance on centralized authorities.

Future enhancements include integration of secure payment gateways, backend identity verification mechanisms, and deployment on a public mainnet environment. Additional improvements may involve role-based access control, automated lease expiration handling, and interoperability with external logistics platforms. These extensions aim to further strengthen AtLease as a scalable and production-ready leasing infrastructure.

REFERENCES

- [1] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.
- [2] G. Wood, "Ethereum: A Secure Decentralised Generalised Transaction Ledger," Ethereum Yellow Paper, 2014.
- [3] V. Buterin, "A Next-Generation Smart Contract and Decentralized Application Platform," Ethereum White Paper, 2014.
- [4] M. Al-Bassam, "Scalable and Secure Asset Transfer via Smart Contracts," in *Proc. IEEE Conf. on Blockchain*, 2018.
- [5] J. Benet, "IPFS - Content Addressed, Versioned, P2P File System," arXiv:1407.3561, 2014.
- [6] E. Entriiken, W. Shirley, J. Evans, and N. Sachs, "ERC-721 Non-Fungible Token Standard," Ethereum Improvement Proposal 721, 2018.
- [7] G. O. Karame and E. Androulaki, "Blockchain-Based Digital Identity Management," *IEEE Security & Privacy*, vol. 15, no. 5, pp. 66–73, 2017.